



Single-shot, single-mask X-ray dark-field, and phase-contrast imaging

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X-ray phase-contrast imaging (XPCI) and dark-field (DF) imaging can potentially provide enhanced contrast beyond attenuation by detecting phase shifts and ultra-small-angle X-ray scattering (USAXS), but most implementations rely on multiple exposures, very high magnification geometry, precision motion, or ultra-high-resolution detectors. In this study, we present a single-mask, single-exposure framework that captures attenuation, differential phase contrast (DPC), and dark-field using mask-pixel alignment technique, which encodes sub-pixel beamlet shifts and broadening into pixel-scale intensity differences on moderate-pitch detectors. We present three configurations that utilize the same hardware: the established single-mask DPC arrangement and two new configurations, DF-only and combined DF-DPC. The desired configuration is selectable by a simple positional adjustment of the mask. We also developed a unified light-transport model that describes signal formation across each of the three configurations and enables accurate single-shot retrieval. Benefiting from the mask-pixel alignment technique and the physics model, our approach eliminates the need for ultra-fine detector resolution as well as mechanical stepping with multiple exposures, thus offering a fast, compact, low-cost, low-dose, and reconfigurable path to advanced multi-contrast X-ray imaging for clinical and industrial applications. These features would also make dark-field and differential phase computed tomography significantly more practical and clinically translatable. © 2025 Optica Publishing Group under the terms of the [Optica Open Access Publishing Agreement](#)

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1. INTRODUCTION

X-ray imaging has long been a fundamental tool in both medical diagnostics and materials science due to its ability to reveal internal structures with high spatial resolution. Traditional X-ray imaging techniques predominantly rely on attenuation contrast, where differences in the absorption of X-rays by various tissue types or materials generate an image. However, this method often falls short when distinguishing between materials with similar attenuation coefficients, especially for light-element materials like soft tissues. Additionally, its ability to detect microstructures is constrained by the resolution limits of conventional imaging systems.

X-ray phase-contrast imaging (XPCI) and dark-field (DF) imaging have emerged as powerful complementary techniques that offer enhanced contrast, particularly for soft tissues and materials with microstructures [1–5]. Phase-contrast imaging is sensitive to the phase shift of X-ray wave that passes through the medium, which exploits variations in the refractive index of different materials at the length scale of tens of micrometers. These variations induce beam refraction at the micro-radian scale [4,5]. This type of technique offers enhanced visualization through signatures like differential phase contrast (DPC) and Laplacian phase signatures, etc.

Dark-field imaging, on the other hand, is sensitive to ultra-small-angle X-ray scattering (USAXS) induced by sub-pixel-scale (microns or sub-micron level) microstructures within the medium. As X-rays pass through, they undergo multiple refractions caused by these microstructures, resulting in USAXS within an extremely narrow angular range [5,6]. This scattering manifests as localized blurring in the images. This phenomenon highlights the presence of microstructures that are invisible in traditional absorption-based imaging, thereby introducing a new dimension of contrast in X-ray imaging.

Most XPCI techniques follow a common principle: a reference intensity pattern is generated and its distortions, caused by refraction and USAXS, are analyzed to retrieve the phase and dark-field signals. Specifically, X-ray attenuation locally reduces the average brightness of the pattern, refraction causes slight local displacements, and scattering causes local blurring, thereby locally decreasing the contrast of the pattern. Therefore, XPCI methods can therefore be classified according to two key aspects: (1) how the reference pattern is generated; and (2) how changes to that pattern are detected.

The detection of reference pattern change remains particularly difficult, since the X-ray refraction angles are typically at micro- to sub-micro-radian range for soft materials. Capturing these effects

often requires ultra-fine detector sampling or multiple steps of data acquisitions with precise movements of the X-ray optics in the beam path.

Propagation-based (PB) imaging [7,8] is the simplest technique for phase-contrast imaging. The method works by increasing the sample-to-detector distance without any additional optics as a reference pattern, allowing phase-contrast patterns to form as the propagation of X-rays. Despite its simplicity, PB imaging requires a high coherence X-ray source and longer exposure times to produce high-quality phase-contrast images. Also, retrieving the phase signal can be challenging, as it usually has extra requirements, like single-material assumption [9], multiple exposures [10], or spectral imaging [11], etc.

Grating interferometry [12,13] typically employs three delicate gratings. The diffraction of the second grating (Talbot effect) forms the reference pattern and is then analyzed by the stepping of the third grating with multiple exposures. The method is compatible with conventional X-ray tubes by incorporating a source grating to create an array of spatially coherent sources. However, its clinical translation faces challenges due to the complexity of the setup, stringent fabrication requirements for gratings with micron-level pitch, and the need for multiple exposures with precise grating movements. Furthermore, the technique is typically optimized for a single wavelength, which limits its performance with broad-spectrum X-ray sources [14–16].

Modulation-based imaging techniques, including speckle-based [17,18], beam-tracking [19–21], and mesh-based methods [22], employ a single optical element to generate a reference intensity pattern. This pattern can arise from either random textures such as sandpaper (speckle-based) or a periodic pattern (beam tracking and mesh-based) [23]. Despite the simplicity, they rely heavily on ultra-high-resolution detectors [17,20,21] or high geometric magnification (e.g., $10\times$ in [19]) to resolve the subtle deformations of the reference pattern caused by refraction and scattering, which remains a significant limitation for translation to clinical and industrial imaging systems.

The edge-illumination (or coded-aperture) method [24] uses two masks (pitch on the order of detector pixel sizes or lower) instead of delicate and high-pitch gratings: the first generates a reference pattern, which is analyzed by the stepping of the second mask with multiple exposures. Unlike grating-based methods, it does not rely on the Talbot effect, making it more tolerant to polychromatic sources. Despite its advantages, it still requires precise alignment, multiple exposures with precise mask movements [25]

or spectral imaging [26] to extract dark-field and differential phase images.

In summary, because detecting subtle changes in the reference pattern is inherently challenging, most of these techniques demand complex and costly setups, often requiring one or more of the following: highly coherent X-ray sources, ultra-high-resolution detectors, sophisticated X-ray gratings, ultra-precise alignment of components, and multiple exposures to retrieve phase-contrast images. These stringent requirements have hindered the broader adoption of these methods in clinical and industrial applications.

The single-mask method, pioneered by Krejci *et al.* [27,28], has emerged as a promising approach to X-ray phase-contrast imaging. This technique requires only one optical element: a periodic absorption mask positioned upstream and close to the sample [Fig. 1(a)]. The mask creates X-ray beamlets [Fig. 1(b)] with the periodic blocking strips made of high-Z material, such as gold. As X-rays pass through the sample, various interactions (attenuation, refraction, and scattering) modify the pattern formed by the mask on the detector plane. As we mentioned before, directly resolving the subtle changes on the reference pattern with the detector requires ultra-fine detector sampling, which is not practical for clinical systems.

To capture the pattern distortions with a lower resolution detector, the single-mask approach employs a different detection principle based on a specialized alignment between the mask and detector pixels, with each pixel effectively acting as an aperture that samples a distinct portion of the beamlet intensity distribution, which allows sub-pixel distortions caused by refraction and scattering to be encoded into measurable intensity variations.

In our recent work, we showed a light-transport model [29] for a single-mask setup that is capable of simultaneously capturing attenuation and differential phase-contrast (DPC) images in a single exposure. This offers high signal-to-noise ratio and dose efficiency in both radiography [30] and phase-contrast micro-CT [31,32].

In this study, we propose two additional variations of the single-mask setup with the capability to obtain dark-field contrast with or without differential-phase images. These three configurations operate on the same hardware platform and can capture attenuation, DPC, and dark-field signals independently and simultaneously from a single-shot acquisition, without the need for multiple exposures, spectral separation, or ultra-high-resolution detectors.

We also propose a unified theoretical framework to describe signal formation across all three configurations, based on the Fokker–Planck equation [33]. While several previous studies have applied transport-of-intensity-based (TIE) models to X-ray phase-contrast imaging, many of them rely on simplifying assumptions, such as a constant attenuation-to-phase ratio throughout the object [9,34,35], or negligible attenuation [36]. Others rely on multiple exposures [37] or spectral decomposition [11,26] for phase retrieval. Also, some of these existing models exclude dark-field contributions entirely. Moreover, most prior TIE-based models assume that the detector fully resolves the wavefield, which usually needs high-resolution detectors or high magnification. In contrast, our model treats attenuation, phase, and dark-field signals as independent variables without those assumptions. Also, we account for sub-pixel beamlet distortions, which enable signal extraction even with limited detector resolution.

Our physics models would allow an intuitive understanding of signal formation in these different configurations. In addition,

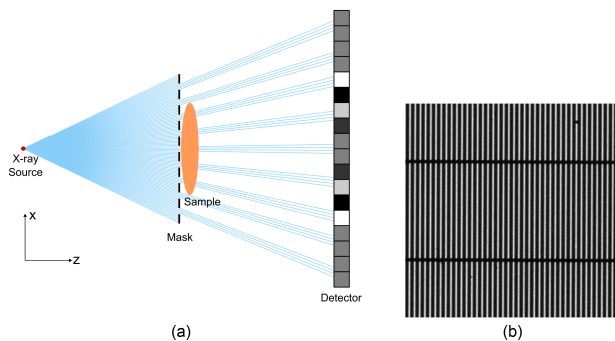


Fig. 1. (a) Diagram for a single-mask X-ray differential phase-contrast imaging system. Figure adapted from our previous publication [29]. (b) Zoomed in high-resolution ($8\ \mu\text{m}$) X-ray image of the mask. The period and opening width of the mask are $53\ \mu\text{m}$ and $20\ \mu\text{m}$, respectively. The horizontal lines are supporting structures of the mask.

it also allows effective retrieval of attenuation, differential phase, and dark-field images from a single exposure without the need for spectral information or complex motion of optical elements. This opens new possibilities for efficient material classification and quantitative imaging across diverse applications, offering practical advantages for both preclinical and potentially clinical translation.

2. LIGHT TRANSPORT MODEL

In our previous work, our formulation for single-mask DPC configuration is based on the approximated transport-of-intensity equation (TIE) [38]. It described the intensity change after the EM wave propagates for a certain distance z :

$$I(\vec{r}, z) = I(\vec{r}, 0) - \frac{z}{k} \nabla_{\perp} \cdot [I(\vec{r}, 0) \nabla_{\perp} \phi(\vec{r}, 0)], \quad (1)$$

where $I(\vec{r}, 0)$ is the X-ray intensity before propagation, $\vec{r} = (x, y)$ is the transverse coordinate, and k is the wave number. The second term describes changes in intensity due to variations of phase shift $\phi(\vec{r}, 0)$ after propagating a distance z , which leads to a redistribution of optical energy, including local convergence/divergence and transverse displacement of the beams.

The X-ray Fokker–Planck equation, proposed by Paganin *et al.* [33], is a modification of TIE that adding a term modeling ultra-small-angle scattering. If we assume the material properties vary slowly within the transverse plane, the equation can be written as

$$I(\vec{r}, z) = I(\vec{r}, 0) - \frac{z}{k} \nabla_{\perp} \cdot [I(\vec{r}, 0) \nabla_{\perp} \phi(\vec{r}, 0)] + z S(\vec{r}) \nabla_{\perp}^2 I(\vec{r}, 0), \quad (2)$$

where $S(\vec{r})$ is defined as the effective scattering coefficient, which denotes the structural property of the sample.

When applying to single-mask imaging system, $I(\vec{r}, 0) = T(\vec{r})M(x)$, where T and M are the transmission functions of the object and the mask, respectively. Similar to our previous work [29], the X-ray intensity measured by a certain detector pixel can be written as the integration within the pixel region. For simplicity, here we focus on a single detector row and I_n is the intensity measured by the n th pixel of this row. Following the process shown in Supplement 1, the equation can be derived as

$$\begin{aligned} I_n = & T(x_n) \int_{x_n}^{x_{n+1}} M(x) dx - \frac{z}{k} T(x_n) \nabla_{\perp}^2 \phi(x_n) \int_{x_n}^{x_{n+1}} M(x) dx \\ & - \frac{z}{k} T(x_n) \partial_x \phi(x_n) \int_{x_n}^{x_{n+1}} M'(x) dx \\ & + z T(x_n) S(x_n) \int_{x_n}^{x_{n+1}} M''(x) dx. \end{aligned} \quad (3)$$

With this unified formulation in place, we will now describe various configurations of single-mask phase imaging that offer enhancement and recovery of different signal types.

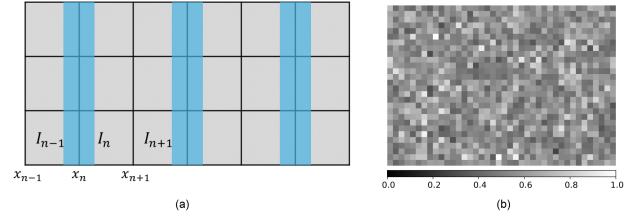


Fig. 2. (a) Diagram of the DPC configuration, showing beamlets (blue) and detector pixels (gray). (b) Zoomed mask image without sample, showing uniform illumination.

3. THREE CONFIGURATIONS OF THE SINGLE-MASK METHOD

A. Single-Mask DPC Configuration

The original single-mask method [27,29] (referred to here as the DPC configuration) positions the mask so that the projected pattern period equals twice the detector pixel size p , with each beamlet aligned to every other pixel boundary [Fig. 2(a)]. While previously published, we present a concise summary here for continuity.

With this alignment, the mask pattern is invisible on the detector without a sample, resulting in uniform pixel intensity [Fig. 2(b)]. Introducing a sample causes beamlet shifts from refraction, creating pixel-level intensity variations detectable even with low-resolution detectors.

The detected intensity at each pixel is given by

$$I_n = w_e T_n (1 - L_n) - \alpha (-1)^n T_n D_n. \quad (4)$$

Here, $T_n = T(x_n)$, $D_n = \frac{z}{k} \partial_x \phi(x_n)$, and $L_n = \frac{z}{k} \nabla_{\perp}^2 \phi(x_n)$ represent the attenuation, differential phase, and Laplacian phase at pixel n , respectively. The coefficients w_e and α correspond to the effective transparency and contrast of the mask (see Supplement 1).

Equation (4) shows that without a sample, the signal is uniform ($T_n = 1$, $L_n = 0$, $D_n = 0$). The presence of a sample introduces a phase gradient D_n , which results in alternating bright and dark fringes.

The signal retrieval equations [29] are

$$\begin{cases} T_n (1 - L_n) \approx \frac{\bar{I}_n + \bar{I}_{n+1}}{2} \\ \frac{\alpha}{w_e} D_n \approx \frac{\bar{I}_n - \bar{I}_{n+1}}{\bar{I}_n + \bar{I}_{n+1}} (-1)^n \end{cases}, \quad (5)$$

where $\bar{I}_n = I_n^{(S)}/I_n^{(M)}$ is the flat-field corrected intensity. The first equation yields the attenuation (with Laplacian enhancement), and the second extracts the DPC signal with system sensitivity proportional to α/w_e .

Figure 3 shows the retrieved attenuation and differential phase images retrieved from a single shot with the single-mask DPC configuration. The sample used was a multi-material phantom, which included a graphite rod with a diameter of 3 mm, several plastic beads with a diameter of 2 mm, and diamond powders with grain size ranging from 50 to 80 μm . All components were placed inside a centrifuge tube. We can see the DPC image demonstrates superior contrast compared to the attenuation image, effectively highlighting the boundaries between various objects.

B. Single-Mask Dark-Field Configuration

Here we present a new configuration, referred to as the dark-field (DF) configuration, which is designed to capture dark-field images. This setup uses the identical X-ray mask and mask-detector

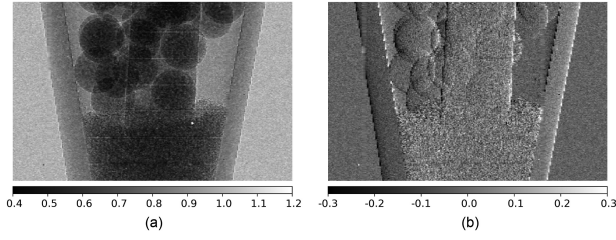


Fig. 3. Retrieved (a) attenuation image and (b) DPC image of the multi-material phantom with single-mask DPC configuration.

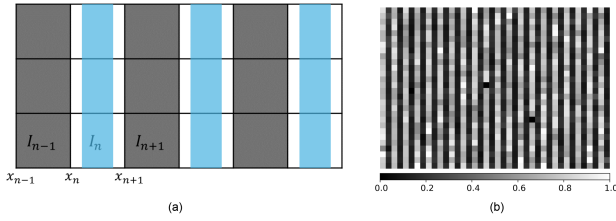


Fig. 4. (a) Diagram of the single-mask dark-field (DF) configuration. The blue strips represent the X-ray beamlets created by the mask, while the squares represent the detector pixels, with their color indicating the X-ray intensity measured by each pixel. (b) Zoomed mask image from the experiment.

pitch relationship to the DPC configuration, in which the period of the projected mask pattern remains twice the pixel size p . The key difference being that the beamlets created by the mask are aligned with the center of every other pixel instead of pixel boundaries, as shown in Fig. 4.

In the absence of a sample, this alignment produces an image with high-contrast bright and dark patterns [Fig. 4(b)]. When a sample is placed in the beam path, some X-ray photons scattered by the sample will strike the dark pixels, decreasing the contrast of the mask pattern locally. Consequently, the dark-field signal can be extracted by comparing the intensity contrast between bright and dark pixels with and without the sample.

Following the process shown in Supplement 1, the intensity received by each pixel can be derived as

$$I_n = [w_e + \alpha_1(-1)^n]T_n(1 - L_n) - \alpha_3(-1)^n T_n S_n, \quad (6)$$

where T_n and L_n represent the same signals as in the previous section, and $S_n = zS(x_n)$ represents the average dark-field signal at the corresponding pixel.

Also, the coefficients in the equation (w_e , α_1 , and α_3) are parameters related to the mask, and the formulas can be found in Supplement 1. w_e is still the equivalent transparency of the mask. Similar to α in Eq. (4), α_1 and α_3 are related to the contrast of the mask.

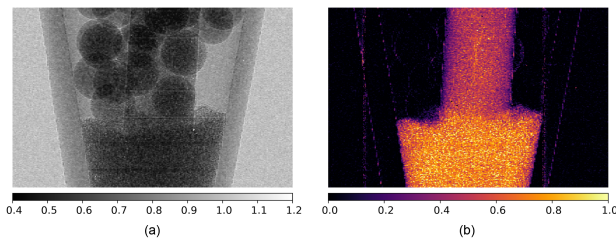


Fig. 5. Retrieved (a) attenuation image and (b) dark-field image of the multi-material phantom with single-mask DF configuration.

From Eq. (6), we can see that without the sample ($T_n = 1$, $L_n = 0$, and $S_n = 0$), the image shows a high-contrast mask pattern with a period of two pixels, characterized by the effective transparency w_e and contrast α_1 . Once the sample is introduced into the beam path, the dark-field signal S_n reduces the local contrast of the mask pattern due to scattering.

As derived in Supplement 1, the retrieval method can be written as

$$\begin{cases} T_n(1 - L_n) = \frac{I_n^{(S)} + I_{n+1}^{(S)}}{I_n^{(M)} + I_{n+1}^{(M)}} \\ \frac{\alpha_3}{\alpha_1} S_n = 1 - \frac{I_n^{(S)} - I_{n+1}^{(S)}}{I_n^{(M)} - I_{n+1}^{(M)}} \cdot \frac{I_n^{(M)} + I_{n+1}^{(M)}}{I_n^{(S)} + I_{n+1}^{(S)}} \end{cases} \quad (7)$$

Similar to the previous section, the right-hand side of the equations is retrieved attenuation (with Laplacian phase enhancement) and dark-field image, respectively. The dark-field intensity ranges from zero to one, indicating the portion of photons scattered to neighboring pixels. In this configuration, the dark-field sensitivity of the imaging system is α_3/α_1 , related to the mask's transmission function.

Figure 5 shows the retrieved attenuation and dark-field images obtained from a single shot using the single-mask DF configuration. The sample is the same as the one used in Fig. 3. From the figure, we can observe that, while the attenuation image [Fig. 5(a)] is capable of depicting various objects, it falls short in distinguishing between smooth, uniform materials, such as the wall of a centrifuge tube and plastic beads, and materials characterized by microstructures, like graphite and diamond powders. In contrast, the dark-field image [Fig. 5(b)] excels in highlighting materials with microstructures, such as graphite and diamond powders, but shows no signal for uniform materials.

However, for capturing differential phase signals, this configuration is ineffective. Since the beamlets are centered on the pixels, micron-level displacements of the beamlet pattern do not alter the total intensity received by each pixel. Thus, this configuration cannot detect differential phase signals.

C. Single-Mask DF-DPC Configuration

Another configuration, termed DF-DPC configuration, combines the features of both DPC and DF configurations. In this setup, the mask should be positioned such that the period of the projected mask pattern is three times the detector pixel size p , with each beamlet aligned with one in every three-pixel boundaries, as illustrated in Fig. 6. Notably, the same mask used in the first two configurations can be employed here; only its position needs adjustment to achieve the required magnification for this specific arrangement.

As a result, the mask-only image exhibits a pattern with a period of three pixels, where two pixels are equally illuminated (bright pixels) and one pixel remains dark [shown in Fig. 6(b)]. This arrangement allows for the extraction of differential phase signals from the two bright pixels, while the dark pixels provide the dark-field signal.

From the same theoretical framework [Eq. (3)], we derive the light transport model for this configuration (with details shown in Supplement 1). The measured intensity by each pixel can be written as

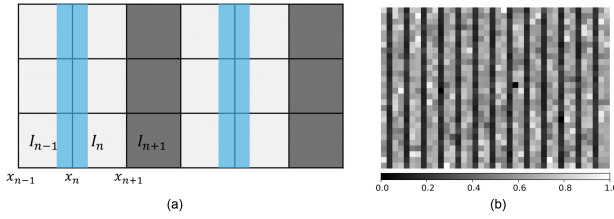


Fig. 6. (a) Diagram of the single-mask dark-field and differential phase-contrast (DF-DPC) configuration. The blue strips represent the X-ray beamlets created by the mask, while the squares represent the detector pixels, with their color indicating the X-ray intensity measured by each pixel. (b) Zoomed mask image from the experiment. This alignment creates a special pattern with three-pixel period.

$$\begin{cases} I_{n-1} = (w_e + \frac{1}{2}\alpha_1) T_{n-1}(1 - L_{n-1}) + \alpha_2 T_{n-1} D_{n-1} - \frac{1}{2}\alpha_3 T_{n-1} S_{n-1} \\ I_n = (w_e + \frac{1}{2}\alpha_1) T_n(1 - L_n) - \alpha_2 T_n D_n - \frac{1}{2}\alpha_3 T_n S_n \\ I_{n+1} = (w_e - \alpha_1) T_{n+1}(1 - L_{n+1}) + \alpha_3 T_{n+1} S_{n+1} \end{cases}, \quad (8)$$

where T_n , D_n , L_n , and S_n represent the attenuation, differential phase, Laplacian phase, and dark-field signal, respectively. Similar to the previous two configurations, the pixel index n corresponds to the position illustrated in Fig. 6(a).

Similar to the previous two configurations, the coefficients in the equation (w_e , α_1 , α_2 , and α_3) are parameters related to the mask and can be found in [Supplement 1](#). w_e is still the equivalent transparency of the mask. α_1 , α_2 , and α_3 are similar to α , α_1 , and α_3 in Eqs. (4) and (6).

The retrieval method can be derived following the process shown in [Supplement 1](#):

$$\begin{cases} T_n(1 - L_n) = \frac{I_{n-1}^{(S)} + I_n^{(S)} + I_{n+1}^{(S)}}{I_{n-1}^{(M)} + I_n^{(M)} + I_{n+1}^{(M)}} \\ \frac{2\alpha_2}{w_e + \frac{1}{2}\alpha_1} D_n = \left(\frac{I_{n-1}^{(S)}}{I_{n-1}^{(M)}} - \frac{I_n^{(S)}}{I_n^{(M)}} \right) \cdot \frac{I_{n-1}^{(M)} + I_n^{(M)} + I_{n+1}^{(M)}}{I_{n-1}^{(S)} + I_n^{(S)} + I_{n+1}^{(S)}} \\ \frac{\alpha_3}{\alpha_1} S_n = \left[1 - \frac{I_{n-1}^{(S)} + I_n^{(S)} - 2I_{n+1}^{(S)}}{I_{n-1}^{(M)} + I_n^{(M)} - 2I_{n+1}^{(M)}} \right] \cdot \frac{I_{n-1}^{(M)} + I_n^{(M)} + I_{n+1}^{(M)}}{I_{n-1}^{(S)} + I_n^{(S)} + I_{n+1}^{(S)}} \end{cases}. \quad (9)$$

The right-hand side of the equations is retrieved attenuation, DPC, and dark-field image, respectively. The DPC and dark-field sensitivity of the imaging system is written as $\frac{2\alpha_2}{w_e + \frac{1}{2}\alpha_1}$ and $\frac{\alpha_3}{\alpha_1}$, respectively.

Figure 7 shows the retrieved attenuation, differential phase, and dark-field images retrieved from a single shot with the single-mask DF-DPC configuration. The sample is the same as the one used in Figs. 3 and 5. We could see the DF-DPC can produce high-quality

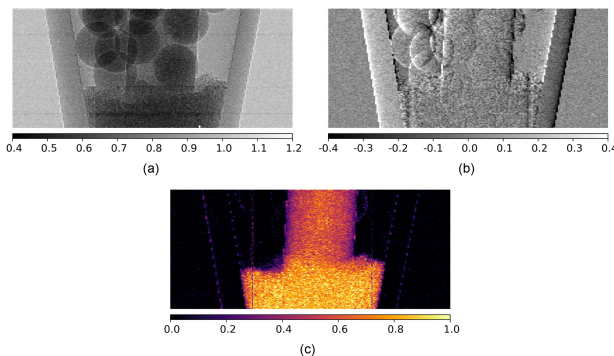


Fig. 7. Retrieved (a) attenuation image, (b) DPC image, and (c) dark-field image of the multi-material phantom with single-mask DF-DPC configuration. (This figure has been presented in the recent SPIE conference [32]).

attenuation, differential phase, and dark-field images similar to the previous configurations.

4. EXPERIMENT

A. X-Ray Source

We utilized a polychromatic micro-focus X-ray tube (Hamamatsu L8121-03) operating with three different focal spot sizes (7, 20, and 50 μm). The tube voltage we used is 40 kV (for non-spectral) and 60 kV (for spectral).

B. Mask

We employed a mask with gold strips, approximately 53 μm in periodicity, fabricated on a graphite substrate. The width of the opening slit of the mask is around 20 μm . The thickness of the gold layer is 100 μm , which can block approximately 91% of X-ray photons with an energy of 40 keV.

C. Detector

The data was collected using a silicon WidePix photon-counting detector [39] with a pixel size of 55 μm and sensor thickness of 500 μm , which was meticulously calibrated and corrected [40,41]. The detector sensor measures approximately 70 mm $\times$ 14 mm, consisting of five identical chips tiled together, with each chip containing 256×256 pixels. Typical exposures yielded 800–1200 photons per pixel, except for the spectral-imaging experiment. For all of the other results shown in the paper including phase retrieval, spectral information achievable from the detector was not used. Our efforts at phase retrieval using non-spectral data acquisition are aimed at easy testing and translation of methods to clinical and industrial settings which use non-spectral (energy-integrating) X-ray detectors which are readily available with larger active area or panel which is required for clinical applications.

D. System Geometry

The source-to-mask distance was fixed at approximately 60 cm, while the mask-to-detector distance varied depending on the configuration:

For DPC or DF configuration, the mask-to-detector distance was approximately 60 cm, resulting in $\sim 2\times$ magnification. This ensured that the period of the projected mask pattern on the detector plane was twice the detector pitch.

For DF-DPC configuration, the mask-to-detector distance was increased to approximately 120 cm, creating $\sim 3\times$ magnification, so the period of the projected mask pattern on the detector plane was three times the detector pitch.

In all configurations, the sample was positioned between the mask and the detector, as close to the mask as possible, to maximize the propagation distance from the sample to the detector.

E. Alignment Method

The alignment between the mask and the detector is achieved through a multi-step process that can potentially be automated. Initially, the mask is placed in an approximate position such that, with the geometric magnification, the period of the projected mask pattern roughly matches the detector pitch. Next, the mask's

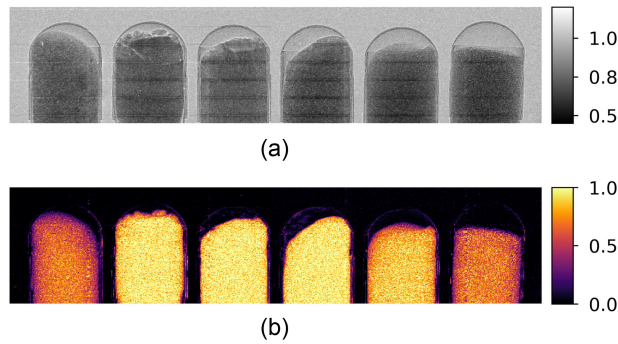


Fig. 8. Retrieved (a) attenuation image and (b) dark-field image of six capsules filled with diamond powders of different grain sizes. From left to right corresponds to the sample numbers 1–6 in Table 1. Images taken with single-mask DF configuration.

Table 1. Approximate Grain Size of Each Diamond Powder Sample

Sample Number	1	2	3	4	5	6
Grain size (μm)	0–0.25	0–0.5	0–1	2–3	10–20	50–80

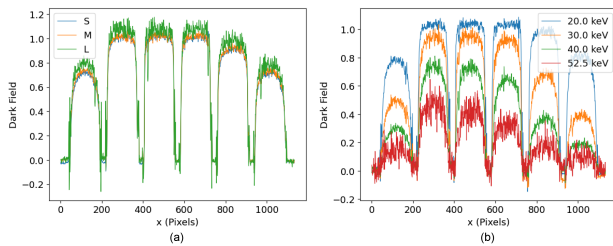


Fig. 9. Cross-sectional profile of retrieved dark-field image of capsules phantom (a) with different focal spot size of the X-ray source: S ($7\ \mu\text{m}$), M ($20\ \mu\text{m}$), and L ($50\ \mu\text{m}$); (b) in different energy windows: 15–25, 25–35, 35–45, and 45–60 keV.

position is fine-tuned using remotely controlled motors. This fine-tuning is guided by observing the Moiré fringes that appear due to slight misalignments. As the alignment improves, the frequency of the Moiré pattern gradually decreases. The process continues until the Moiré pattern disappears, indicating that the period of the projected mask pattern is well matched with the detector pitch. Finally, the mask is moved horizontally until it reaches the precise position required for the chosen configuration.

5. RESULTS

To investigate the dark-field sensitivity of our imaging system over different micro-structure scales, we used a phantom consisting of six capsules filled with diamond powders of varying grain sizes. This is a common strategy used in several previous studies for existing dark-field imaging methods [42–47]. The retrieved attenuation and dark-field images using the single-mask DF configuration are shown in Fig. 8. The grain sizes of the diamond powders in each capsule are shown in Table 1. The measurements were taken with three different focal spot sizes of the X-ray source (S: $7\ \mu\text{m}$, M: $20\ \mu\text{m}$, and L: $50\ \mu\text{m}$) and across different energy windows (15–25, 25–35, 35–45, and 45–60 keV). The cross-sectional plots are shown in Fig. 9.

From the attenuation image in Fig. 8(a), we can barely see any differences between the six capsules since their infills have almost the same chemical composition. However, the dark-field image [Fig. 8(b)] shows significant differences between each sample. We observe that Samples 2, 3, and 4 exhibit the strongest dark-field signals, indicating that our X-ray dark-field imaging system has the highest sensitivity to features with sizes ranging from 0.5 to $3\ \mu\text{m}$.

Figure 9(a) shows that within the range up to $50\ \mu\text{m}$, a larger focal spot size does not impact the contrast of the dark-field signal. In contrast, according to our previous study [30], the contrast of DPC images decreases as the focal spot size increases. Additionally, Fig. 9(b) shows the profile plot in several different energy windows. From these results, we can see the imaging system works under a wide spectrum. We also notice the decrease in dark-field signal with higher X-ray energies, which is within our expectation since X-ray refraction is weaker with higher X-ray energy. Overall, the sensitivity response to different structure sizes as well as the energy dependencies is consistent with behaviors previously reported in grating-based XPCI systems [6].

Figure 10 showcases the retrieved images of a dried fish sample using a single shot with the single-mask DF configuration. The attenuation image [Fig. 10(a)] provides an overall view of the sample's structure, whereas the dark-field image [Fig. 10(b)] reveals some special features that are not visible in the attenuation image. The combined image [Fig. 10(c)] merges information from both modalities, offering a comprehensive view of the sample, highlighting both structural and microstructural details. These images showed the efficacy of dark-field imaging in capturing detailed structural information that is otherwise undetectable with conventional absorption imaging.

6. DISCUSSION

The single-mask phase-contrast and dark-field imaging approach offers practical advantages in terms of system simplicity and flexibility, while also presenting unique challenges and trade-offs. Below, we discuss several important aspects that influence system performance and clinical applicability.

One of the core challenges in XPCI is to detect micron-scale refraction and USAXS signals. Grating-based and edge-illumination techniques use phase stepping, requiring multiple exposures with sub-pixel shifts for signal extraction. Speckle-based, beam-tracking, and mesh methods use high-resolution detectors [17,20,21] or large magnification [19] to directly resolve deformations in reference patterns. In contrast, our method adopts a third strategy based on mask-to-pixel alignment. By precisely aligning the mask pattern with the detector pixels, the system detects sub-pixel distortions in beamlet intensity distributions without requiring high-resolution detectors or multiple exposures. This detection approach offers a practical balance between sensitivity, resolution, and system simplicity.

In general, the performance of phase-contrast and dark-field imaging systems depends critically on the spatial coherence of the X-ray beam, which in turn is influenced by the source focal spot size. Our experiments demonstrate robust performance at focal spot sizes up to $50\ \mu\text{m}$, the largest available on our system. This tolerance arises from the geometric configuration of our single-mask setup. However, we acknowledge that at much larger spot sizes (e.g., in millimeter level typical of clinical radiography), the beamlet profiles would broaden significantly, blurring the mask pattern on the detector plane and ultimately the system's ability to

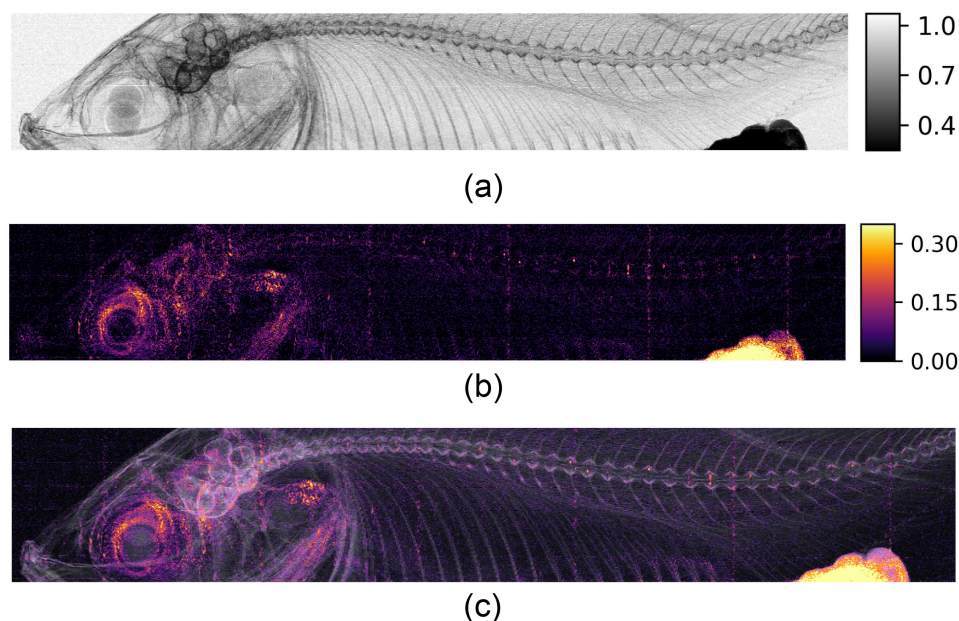


Fig. 10. Retrieved images of a dried fish from single-mask DF configuration. (a) Attenuation image; (b) dark-field image; and (c) image combining attenuation and dark-field signal.

produce usable contrast. In such cases, introducing a source mask, similar to the G0 grating in Talbot–Lau interferometry, could restore spatial coherence. Our results provide a practical reference: an aperture width of $\sim 50\ \mu\text{m}$ remains compatible with contrast preservation in this configuration.

Our experimental results demonstrate that with our current setup, the single-mask system is capable of detecting USAXS signals from a range of microstructures, from sub-micrometer level to $80\ \mu\text{m}$ level, which is the largest diamond powder sample available in our lab. We show that using a simple setup, in a single shot, without high-resolution detectors, we can extract dark-field signal from structures of similar size ranges shown by existing techniques [42–47]. Theoretically, a smaller mask or grating pitch would improve phase and dark-field sensitivity of both grating-based, beam-tracking, and our method. A direct quantitative comparison with other methods is not within the scope of this paper, but would be interesting to do as part of future work.

When comparing the three configurations, our results (Figs. 3, 5, and 7) show that the DF-DPC setup produced slightly higher contrast in the DPC and DF images, due to its longer effective propagation distance. However, it should also be noted that the DF–DPC configuration may exhibit reduced sensitivity to weakly scattering materials, because the distance between the beamlets and the dark pixels is larger in this configuration. As a result, scattered beamlets must deviate by a greater angle to reach adjacent dark pixels, which slightly raises the detection threshold. This angular acceptance effect is a trade-off of this configuration and worth further investigation in future studies.

Another aspect worth noting is that, for all XPCI systems utilizing a reference pattern to extract phase and dark-field signals, the effective image resolution is limited not only by the detector pixel size but also by the spatial period of the reference pattern, whichever is larger. In our case, the mask period corresponds to two pixels for the DPC or DF configurations, and three pixels for the DF–DPC configuration, which results in a slight reduction in

spatial resolution compared with the detector’s native sampling, a common characteristic of this type of technique.

An advantage of our imaging system is its compatibility with polychromatic X-ray spectra, unlike grating-based methods that rely on the energy-dependent Talbot effect. In such systems, fringe visibility degrades under broad-spectrum illumination with typical visibility around 20% even after optimization [48]. In our setup, beamlet formation is governed primarily by geometric projection rather than interference, avoiding visibility loss under a broad energy range. As a result, it can achieve higher visibility (near 50%), with a gradual decline at higher energies due to the reduced attenuation of gold. Although the forward model (based on Fokker–Planck equation) is energy-dependent, the polychromatic measurements can be considered as effective-energy-weighted integrations, which is sufficient for many imaging tasks.

Translating this technique to the scale of clinical imaging (like chest radiography) requires consideration of tolerance to geometric misalignments. While our approach requires alignment of the mask with the detector at the pixel level, fortunately, the alignment tolerance in our system is on the order of the aperture width ($\sim 10\ \mu\text{m}$), which we believe is achievable with modern manufacturing and assembly techniques. Additionally, our method uses a relatively larger mask pitch, and the number of required mask elements is lower than that of grating-based systems, which may simplify large-area fabrication and alignment. Future work will explore tile-based mask designs and automated alignment procedures to enable deployment in large-area detector arrays.

7. CONCLUSION

The single-mask X-ray imaging setup presented in this study represents a significant advancement in multi-contrast imaging, providing a practical and efficient method for simultaneously capturing attenuation, differential phase-contrast (DPC), and dark-field images in a single exposure. By utilizing the specially designed mask-pixel alignment, this technique avoids the need

for ultra-high-resolution detectors, phase stepping, or intricately fabricated gratings, which would significantly enhance the field of X-ray phase-contrast imaging with simple, compact, low-dose, and practical methods. These factors, particularly the single-shot, low-dose as well as low-resolution detectors used in our method, would make clinical translation to yield computed tomographic imaging of X-ray dark-field and differential-phase imaging significantly more faster, dose efficient, and practical.

With the additional two configurations proposed, the three configurations of this technique offer exceptional flexibility for different specific applications: whether emphasizing phase-contrast, dark-field signals, or a combination of both. Notably, transitioning between these variations requires no replacement of equipment; only a simple adjustment in the positioning of optical elements is needed. This adaptability enables the achievement of all three imaging modalities within a single system, enhancing its versatility and utility across diverse applications.

Our preliminary investigations show a single-shot dark-field retrieval from a range of structure sizes from 250 nm to 80 μm with a relatively low resolution detector of 55 μm . System design can be optimized to get dark-field images from much smaller or larger structure sizes based on the application and would be part of future work. Signal detection benefits when both differential phase and dark-field signatures are available would also be part of future work.

Building on our previous work, the light-transport model developed for these three variations provides a clear and intuitive framework for understanding signal formation and offers robust retrieval methods for each contrast type. The model also elucidates the influence of system parameters on the sensitivity of each imaging modality, guiding further optimization and application. Future work will focus on investigating and analyzing the sensitivity of various retrieved features to key design parameters.

Overall, this work lays the groundwork for more accessible and cost-effective X-ray imaging solutions with broad potential applications in clinical diagnostics and industrial inspection. The innovations presented here promise to enhance the capabilities of X-ray imaging systems, making advanced multi-contrast imaging a viable option for a wider range of settings and users.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

REFERENCES

- R. Fitzgerald, "Phase-sensitive X-Ray imaging," *Phys. Today* **53**, 23–26 (2000).
- R. A. Lewis, "Medical phase contrast x-ray imaging: current status and future prospects," *Phys. Med. Biol.* **49**, 3573 (2004).
- S. D. Auweter, J. Herzen, M. Willner, *et al.*, "X-ray phase-contrast imaging of the breast—advances towards clinical implementation," *Br. J. Radiol.* **87**, 20130606 (2014).
- S. Tao, C. He, X. Hao, *et al.*, "Principles of different X-ray phase-contrast imaging: a review," *Appl. Sci.* **11**, 2971 (2021).
- L. Quenot, S. Bohic, and E. Brun, "X-ray phase contrast imaging from synchrotron to conventional sources: a review of the existing techniques for biological applications," *Appl. Sci.* **12**, 9539 (2022).
- S. K. Lynch, V. Pai, J. Auxier, *et al.*, "Interpretation of dark-field contrast and particle-size selectivity in grating interferometers," *Appl. Opt.* **50**, 4310–4319 (2011).
- S. W. Wilkins, T. E. Gureyev, D. Gao, *et al.*, "Phase-contrast imaging using polychromatic hard X-rays," *Nature* **384**, 335–338 (1996).
- Y. Suzuki, N. Yagi, and K. Uesugi, "X-ray refraction-enhanced imaging and a method for phase retrieval for a simple object," *J. Synchrotron Radiat.* **9**, 160–165 (2002).
- D. Paganin, S. C. Mayo, T. E. Gureyev, *et al.*, "Simultaneous phase and amplitude extraction from a single defocused image of a homogeneous object," *J. Microsc.* **206**, 33–40 (2002).
- A. Yan, X. Wu, and H. Liu, "An attenuation-partition based iterative phase retrieval algorithm for in-line phase-contrast imaging," *Opt. Express* **16**, 13330–13341 (2008).
- D. Gürsoy and M. Das, "Single-step absorption and phase retrieval with polychromatic x rays using a spectral detector," *Opt. Lett.* **38**, 1461–1463 (2013).
- T. Weitkamp, A. Diaz, C. David, *et al.*, "X-ray phase imaging with a grating interferometer," *Opt. Express* **13**, 6296–6304 (2005).
- F. Pfeiffer, T. Weitkamp, O. Bunk, *et al.*, "Phase retrieval and differential phase-contrast imaging with low-brilliance X-ray sources," *Nat. Phys.* **2**, 258–261 (2006).
- K. Taphorn, F. De Marco, J. Andrejewski, *et al.*, "Grating-based spectral X-ray dark-field imaging for correlation with structural size measures," *Sci. Rep.* **10**, 13195 (2020).
- X. Zhu, C. Wu, H. Gao, *et al.*, "Visibility of polychromatic grating-based X-ray imaging systems based on wave-optical theories and frequency domain analysis," *Front. Phys.* **9**, 630337 (2021).
- G. Pelzer, A. Zang, G. Anton, *et al.*, "Energy weighted x-ray dark-field imaging," *Opt. Express* **22**, 24507–24515 (2014).
- K. S. Morgan, D. M. Paganin, and K. K. W. Siu, "X-ray phase imaging with a paper analyzer," *Appl. Phys. Lett.* **100**, 124102 (2012).
- M.-C. Zdora, "State of the art of X-ray speckle-based phase-contrast and dark-field imaging," *J. Imaging* **4**, 60 (2018).
- F. A. Vittoria, G. K. N. Kallon, D. Basta, *et al.*, "Beam tracking approach for single-shot retrieval of absorption, refraction, and dark-field signals with laboratory x-ray sources," *Appl. Phys. Lett.* **106**, 224102 (2015).
- F. A. Vittoria, M. Endrizzi, G. K. Kallon, *et al.*, "Multimodal phase-based X-ray microtomography with nonmicrofocal laboratory sources," *Phys. Rev. Appl.* **8**, 064009 (2017).
- A. Pil-Ali, S. Q. Z. Shi, S. Adnani, *et al.*, "Multimodal single-shot x-ray phase-contrast imaging using a high-resolution micropillar-based grating," *Proc. SPIE* **12463**, 1246309 (2023).
- J. C. Petrucci, D. Hayden, S. Starr-Baier, *et al.*, "Phase and coherent scatter imaging for improved discrimination of low-density materials," *Proc. SPIE* **9847**, 98470C (2016).
- M. A. Beltran, D. M. Paganin, M. K. Croughan, *et al.*, "Fast implicit diffusive dark-field retrieval for single-exposure, single-mask x-ray imaging," *Optica* **10**, 422–429 (2023).
- A. Olivo and R. Speller, "A coded-aperture technique allowing x-ray phase contrast imaging with conventional sources," *Appl. Phys. Lett.* **91**, 074106 (2007).
- A. Olivo, "Edge-illumination x-ray phase-contrast imaging," *J. Phys. Condens. Matter* **33**, 363002 (2021).
- M. Das and Z. Liang, "Spectral x-ray phase contrast imaging for single-shot retrieval of absorption, phase, and differential-phase imagery," *Opt. Lett.* **39**, 6343 (2014).
- F. Krejci, J. Jakubek, and M. Kroupa, "Hard x-ray phase contrast imaging using single absorption grating and hybrid semiconductor pixel detector," *Rev. Sci. Instrum.* **81**, 113702 (2010).
- D. Vavrik, J. Jakubek, I. Jandajsek, *et al.*, "Visualization of delamination in composite materials utilizing advanced X-ray imaging techniques," *J. Instrum.* **10**, C04012 (2015).
- J. Yuan and M. Das, "Transport-of-intensity model for single-mask x-ray differential phase contrast imaging," *Optica* **11**, 478–484 (2024).
- J. Yuan, I. Harmon, and M. Das, "Comparing SNR benefits of single-mask edge-illumination and free-space propagation x-ray phase contrast imaging," *Proc. SPIE* **PC12031**, PC120310X (2022).

31. J. Yuan, I. Harmon, J. C. R. Luna, *et al.*, "Single-mask phase imaging micro-computed tomography with multi-contrast images and single-shot per projection angle," *Proc. SPIE* **12925**, 129253R (2024).
32. J. Yuan, J. C. R. Luna, and M. Das, "Single-shot dark-field and phase contrast micro-CT with single-mask phase contrast imaging," *Proc. SPIE* **13405**, 134051H (2025).
33. D. M. Paganin and K. S. Morgan, "X-ray Fokker-Planck equation for paraxial imaging," *Sci. Rep.* **9**, 17537 (2019).
34. P. C. Diémoz, F. A. Vittoria, C. K. Hagen, *et al.*, "A single-image retrieval method for edge illumination X-ray phase-contrast imaging: application and noise analysis," *Phys. Medica* **32**, 1759–1764 (2016).
35. K. M. Pavlov, H. T. Li, D. M. Paganin, *et al.*, "Single-shot X-ray speckle-based imaging of a single-material object," *Phys. Rev. Appl.* **13**, 054023 (2020).
36. D. M. Paganin, H. Labriet, E. Brun, *et al.*, "Single-image geometric-flow x-ray speckle tracking," *Phys. Rev. A* **98**, 053813 (2018).
37. L. Quénot, H. Rougé-Labriet, S. Bohic, *et al.*, "Implicit tracking approach for X-ray phase-contrast imaging with a random mask and a conventional system," *Optica* **8**, 1412–1415 (2021).
38. C. Zuo, J. Li, J. Sun, *et al.*, "Transport of intensity equation: a tutorial," *Opt. Lasers Eng.* **135**, 106187 (2020).
39. R. Ballabriga, J. Alozy, G. Blaj, *et al.*, "The Medipix3RX: a high resolution, zero dead-time pixel detector readout chip allowing spectroscopic imaging," *J. Instrum.* **8**, C02016 (2013).
40. M. Das, B. Kandel, C. S. Park, *et al.*, "Energy calibration of photon counting detectors using x-ray tube potential as a reference for material decomposition applications," *Proc. SPIE* **9412**, 941214 (2015).
41. S. Vespucci, C. S. Park, R. Torrico, *et al.*, "Robust energy calibration technique for photon counting spectral detectors," *IEEE Trans. Med. Imaging* **38**, 968–978 (2019).
42. M. Kagias, Z. Wang, K. Jefimovs, *et al.*, "Dual phase grating interferometer for tunable dark-field sensitivity," *Appl. Phys. Lett.* **110**, 014105 (2017).
43. A. Pandeshwar, M. Kagias, Z. Wang, *et al.*, "Modeling of beam hardening effects in a dual-phase X-ray grating interferometer for quantitative dark-field imaging," *Opt. Express* **28**, 19187–19204 (2020).
44. S. Lee, O. Oh, Y. Kim, *et al.*, "Study on dark-field imaging with a laboratory x-ray source: Random stress variation analysis based on x-ray grating interferometry," *Rev. Sci. Instrum.* **92**, 015103 (2021).
45. S. Spindler, D. Etter, M. Rawlik, *et al.*, "The choice of an autocorrelation length in dark-field lung imaging," *Sci. Rep.* **13**, 2731 (2023).
46. S. Meyer, S. Z. Shi, N. Shapira, *et al.*, "Quantitative analysis of speckle-based X-ray dark-field imaging using numerical wave-optics simulations," *Sci. Rep.* **11**, 16113 (2021).
47. Y. Y. How, D. M. Paganin, and K. S. Morgan, "On the quantification of sample microstructure using single-exposure x-ray dark-field imaging via a single-grid setup," *Sci. Rep.* **13**, 11001 (2023).
48. M. Viermetz, N. Gustschin, C. Schmid, *et al.*, "Dark-field computed tomography reaches the human scale," *Proc. Natl. Acad. Sci.* **119**, e2118799119 (2022).